THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
MATH3603 Probability Theory

Tutorial 11 Solution
2026/04/15 and 2026/04/17

In this tutorial, we use p_{ij}^n to denote the probability $P(X_n = j \mid X_0 = i)$. Please distinguish it from the n-th power of p_{ij}, which will be denoted as $(p_{ij})^n$.

1. Prove that any state of a finite irreducible Markov Chain is recurrent.

   Solution: It suffices to show that there is at least one recurrent state (which implies that all states are recurrent by irreducibility). Assume to the contrary that all states are transient. Label the states as $\{1, \ldots, n\}$, let N_i be the number of times the process visits state i, and define $N := \sum_{i=1}^n N_i$. We have
   $$\mathbb{E}[N_i \mid X_0 = i] < \infty.$$
   and
   $$\mathbb{E}[N_i \mid X_0 = j] \leq \mathbb{E}[N_i \mid X_0 = i] < \infty \quad \text{for all } j \neq i.$$
   Thus $\mathbb{E}[N_i] < \infty$ for any distribution of X_0, and hence $\mathbb{E}[N] = \sum_{i=1}^n \mathbb{E}[N_i] < \infty$. Since N is non-negative integer-valued, we have $\mathbb{E}[N] = \sum_{n=0}^\infty P(N > n)$, which implies
   $$P(N > n) \to 0 \quad \text{as } n \to \infty.$$
   On the other hand, by the definition of N, we have $N \geq \sum_{j=1}^m \mathbf{1}_{\{X_j \in \{1,\ldots,n\}\}} = m$ for all m, which implies $P(N > n) = 1$ for all n, a contradiction.

2. Show that if state i is recurrent and state i does not communicate with state j, then $p_{ij} = 0$. This implies that once a progress enters a recurrent class of states it can never leave that class. For this reason, a recurrent class is often referred to as a *closed class*.

   Solution: Denote the state space as S. For any states $k_1, k_2 \in S$, define
   $$f_{k_1 k_2} = P(X_n = k_2 \text{ for some } n > 0 \mid X_0 = k_1).$$
   Clearly $i \neq j$ by assumption, and hence $p_{ji}^0 = 0$. Assume to the contrary that $p_{ij} > 0$. Hence $i \to j$. Since state i does not communicate with state j, state i is NOT accessible from state j. i.e. $f_{ji} = 0$. Conditioning on the first step of the Markov chain, we have
   $$\begin{aligned}
   f_{ii} &= P(X_n = i \text{ for some } n > 0 \mid X_0 = i) \\
   &= \sum_{k \in S} P(X_1 = k \mid X_0 = i) P(X_n = i \text{ for some } n > 0 \mid X_0 = i, X_1 = k) \\
   &= p_{ii} + \sum_{k \in S \setminus \{i\}} P(X_1 = k \mid X_0 = i) P(X_n = i \text{ for some } n > 0 \mid X_1 = k) \\
   &= p_{ii} + p_{ij} f_{ji} + \sum_{k \in S \setminus \{i,j\}} p_{ik} f_{ki} \\
   &\leq p_{ii} + \sum_{k \in S \setminus \{i,j\}} p_{ik} = 1 - p_{ij} \\
   &< 1,
   \end{aligned}$$

where the third equality follows from the Markov property, and the first inequality follows from $f_{ji} = 0$ and $f_{ki} \leq 1$ for all k. This contradicts the assumption that state i is recurrent (which implies $f_{ii} = 1$).

3. Let P be the transition probability matrix of a finite-state Markov chain. Prove that if the entries of P^r are all positive for some $r > 0$, then for any integer $n \geq r$, the entries of P^n are all positive.

   Solution: Denote S as the state space. Since every entry of P^n is nonnegative. we have $p_{ij}^n \geq 0$ for all states i, j. For all $n > r$, by the Chapman-Kolmogorov equations we have

   $$p_{ij}^n = \sum_{k \in S} p_{ik}^{n-r} p_{kj}^r.$$

   Since p_{kj}^r is just the (k, j)-entry of P^r, we have $p_{kj}^r > 0$ for all j, k. By $\sum_{k \in S} p_{ik}^{n-r} = 1$, there exists at least one k such that $p_{ik}^{n-r} > 0$. Then $p_{ik}^{n-r} p_{kj}^r > 0$ and thus $p_{ij}^n \geq p_{ik}^{n-r} p_{kj}^r > 0$.

4. Specify the classes of the Markov chains corresponding to the following probability transition matrices, and determine whether each class is transient or recurrent:

   $$P_1 = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix} \qquad P_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

   $$P_3 = \begin{bmatrix} 1/2 & 0 & 1/2 & 0 & 0 \\ 1/4 & 1/2 & 1/4 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 0 & 1/2 & 1/2 \end{bmatrix} \quad P_4 = \begin{bmatrix} 1/4 & 3/4 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

   Solution:

   Without loss of generality, we label the states as $\{0, 1, \ldots\}$.

   P_1: Since $p_{ij} > 0$ for all $i \neq j$, each state communicates with all states (recall that $i \leftrightarrow i$ always). Thus all states are in one class, which is recurrent by the result of Question 1. More precisely, one can show that $p_{ii}^n = \frac{1}{3} + \frac{2}{3}\left(\frac{-1}{2}\right)^n$ for every state i and time n.

   P_2: State 0 and 1 always go to State 3, State 3 always go to State 2 and State 2 can go either State 0 or 1. So all states communicate and thus by Q1 they are all recurrent.

   P_3: State 0 and State 2 belong to one closed class, State 3 and State 4 belong to another closed class. So State 0,2,3,4 are recurrent. Since State 1 has a chance to move to the class of State 0 and it is recurrent, according to Q2 it won't leave the class and thus State 1 is transient.

   P_4: Again State 0 and State 1 is in one recurrent class. State 2 don't go to other state so is also recurrent. State 3 has 1/3 chance to go to State 2 and never come back so it is transient. Finally State 4 always goes to State 1 and never come back so it is also transient.

5. A DNA nucleotide can take one of the four values in $\{A, C, G, T\}$. A standard model for nucleotide mutational change is given by a Markov chain. Specifically, in each time step, each nucleotide does not change with probability $1 - 3\alpha$ for some $0 < \alpha < 1/3$, or it mutates to one

of the other 3 values with equal probability. Find the probability that a given nucleotide is the same as its initial value after n time steps.

Solution: We have the transition matrix for the Markov chain as

$$P = \begin{bmatrix} 1-3\alpha & \alpha & \alpha & \alpha \\ \alpha & 1-3\alpha & \alpha & \alpha \\ \alpha & \alpha & 1-3\alpha & \alpha \\ \alpha & \alpha & \alpha & 1-3\alpha \end{bmatrix} = (1-4\alpha)I_4 + \alpha J_4.$$

where I_4 is the 4×4 identity matrix and J_4 is the 4×4 matrix that every entry is 1. Note that $(J_4)^n = 4^{n-1}J_4$. It follows that

$$\begin{aligned} P^n &= ((1-4\alpha)I_4 + \alpha J_4)^n \\ &= (1-4\alpha)^n I_4 + \sum_{k=1}^{n} \binom{n}{k} (1-4\alpha)^{n-k} \alpha^k (J_4)^k \\ &= (1-4\alpha)^n I_4 + \left(\sum_{k=1}^{n} \binom{n}{k} (1-4\alpha)^{n-k} \alpha^k 4^{k-1} \right) J_4 \\ &= (1-4\alpha)^n I_4 + \frac{1}{4} \left(\sum_{k=1}^{n} \binom{n}{k} (1-4\alpha)^{n-k} (4\alpha)^k \right) J_4 \\ &= (1-4\alpha)^n \left(I_4 - \frac{1}{4} J_4 \right) + \frac{1}{4} \left(\sum_{k=0}^{n} \binom{n}{k} (1-4\alpha)^{n-k} (4\alpha)^k \right) J_4 \\ &= (1-4\alpha)^n \left(I_4 - \frac{1}{4} J_4 \right) + \frac{1}{4} (1-4\alpha+4\alpha)^n J_4 \\ &= (1-4\alpha)^n \left(I_4 - \frac{1}{4} J_4 \right) + \frac{1}{4} J_4. \end{aligned}$$

Thus, the desired probability is given by p_{ii}^n, which equals

$$\begin{aligned} P_{ii}^n &= (1-4\alpha)^n \left(1 - \frac{1}{4} \right) + \frac{1}{4} \\ &= \frac{1}{4} + \frac{3}{4} (1-4\alpha)^n. \end{aligned}$$